



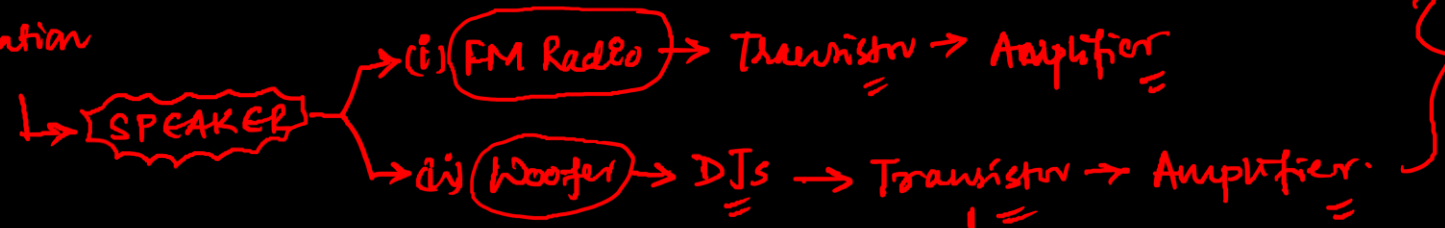
VLSI DESIGN FLOW | 26-07-2025

Er. Sailesh Mishra

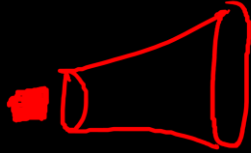
B.Tech (ECE | Silicon University)

Specialization in Microelectronics & VLSI

Application



"AMPLIFIER" is a Core Analog Block.

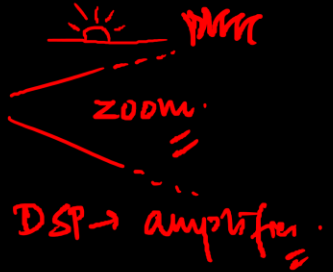
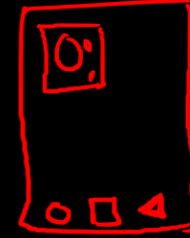
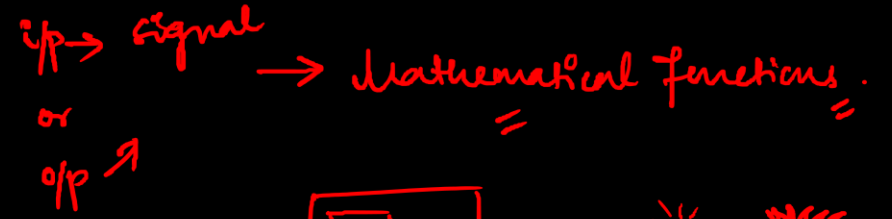


FM Radio :-

- BW ✓ = Band of frequencies.
Gain $\rightarrow \infty$

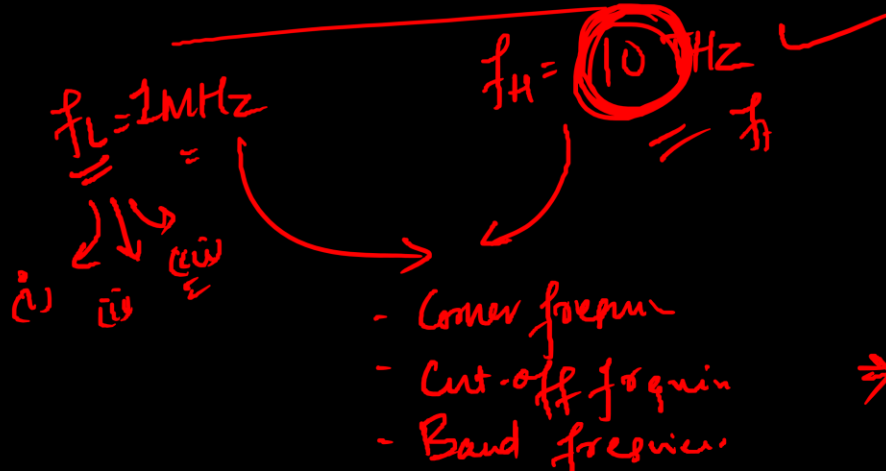
Speaker.

- Gain ✓
40dB
30dB



93.5 Hz

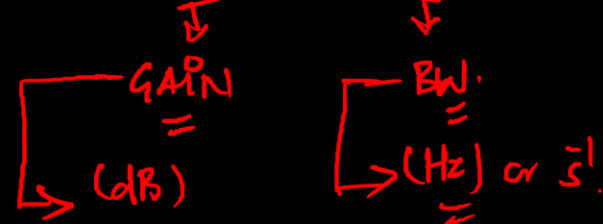
1 MHz ✓



AMPLIFIER

$\rightarrow$ FM Radio = BW, Gain ✓

$\rightarrow$ Speaker = BW, Gain ✓



$\Rightarrow 20 \log(A_v) \rightarrow \frac{V_o}{V_{in}}$

Complex H/W
very costly

Backend design

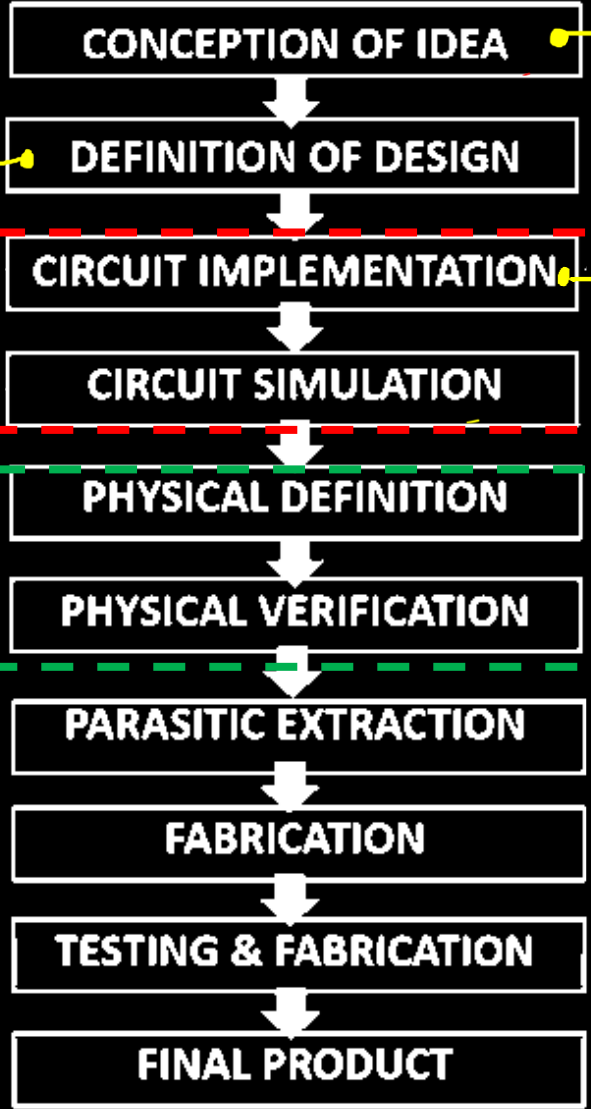
FULL CUSTOM DESIGN

SEMI-CUSTOM DESIGN

Methodology or Approach

Less H/W complex
FRONTEND design

IC DESIGN FLOW



CLIENT'S segment of products =
OLED - laptop...
Smart phone
Camera
FM Radio
Speaker
Calculator

CLIENT'S segment of products =

BTs ✓
MOSFETS ✓

Transistors ✓

DEVICES to be used for the Project

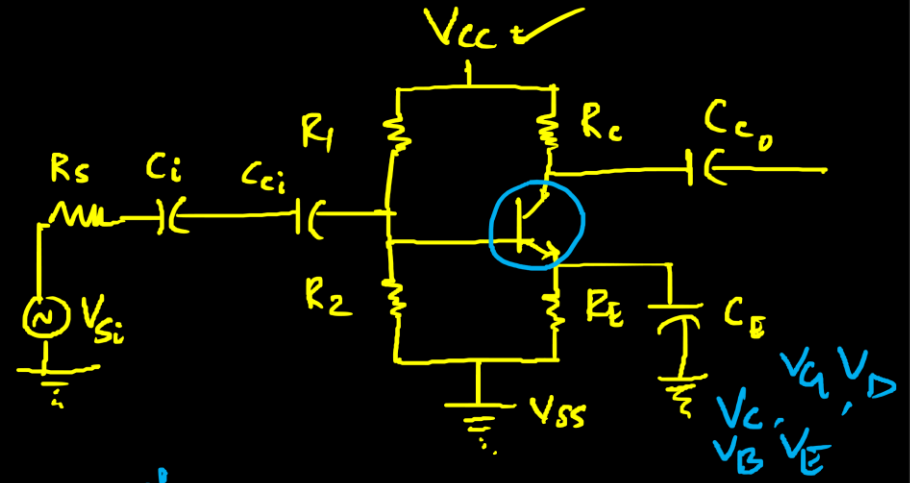
R. → Fabrication process (Layers)
C → (Layers)

Heat sink
Power Transistor =

Fabrication process (Layers)

LAYOUT of the ckt.

M1
M2
Layout =



(i) BJT/MOSFET

(ii) BIASING :-

- g_m ✓
- Noise Margin ✓
- Z_i, Z_o ✓
- Gain ✓
- BWT ✓

- Voltage divider
- Emitter follower
- Fixed Bias
- Stability Analysis.

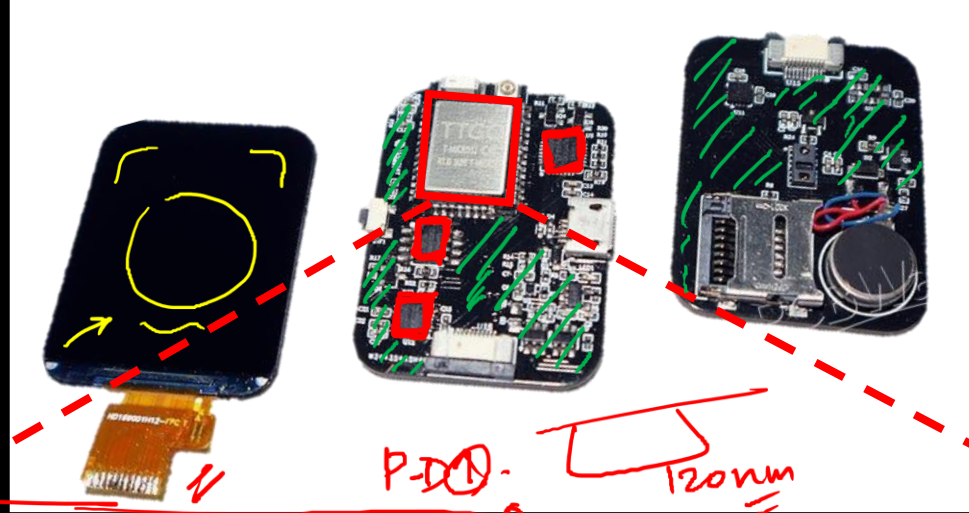
VLSI DESIGN FLOW (Contd.) | 30-07-2025

Er. Sailesh Mishra

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Specialization in Microelectronics & VLSI

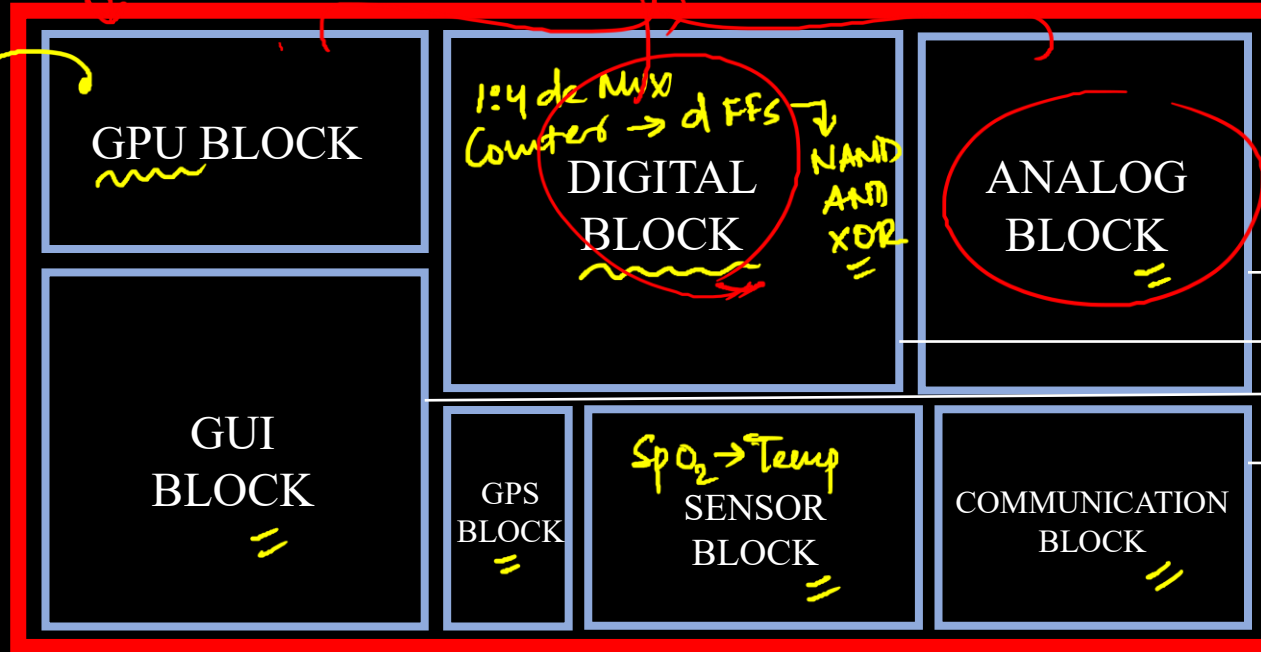
I. SYSTEM SPECIFICATIONS



- MODULARITY. ✓✓
- REGULARITY. ✓✓
- LOCALITY. ✓✓

Graphical
processing
Unit
(GPU)

II. ARCHITECTURAL DESIGN



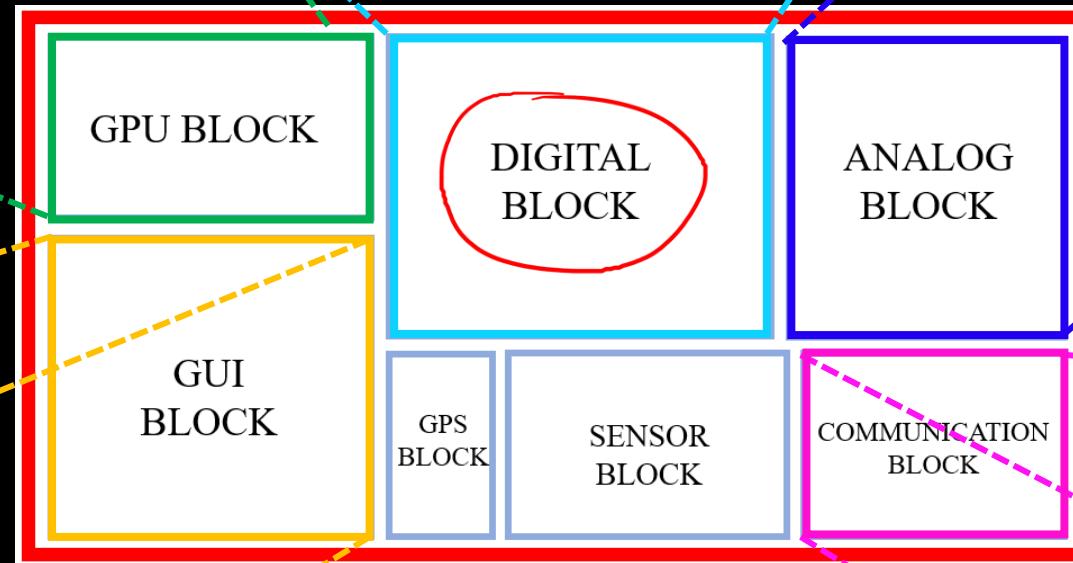
90nm
= P.D. Ⓢ

SUB
BLOCKS

- I. Geometry Processor
- II. Rasterizer
- III. Texture Mapping Unit (TMU)
- IV. Shader Core
- V. Framebuffer Controller
- VI. Display Interface
- VII. 2D/3D Acceleration Engine
- VIII. Video Decoder
- IX. Command Processor
- X. Memory Interface (DRAM Controller)

- I. System Clock Generator
- II. Power Management Controller
- III. Task Scheduler
- IV. Interrupt Controller
- V. Real-Time Clock (RTC)
- VI. Boot ROM
- VII. Internal SRAM
- VIII. Cache Controller
- IX. Memory Protection Unit (MPU)
- X. Debug Interface (JTAG/SWD)

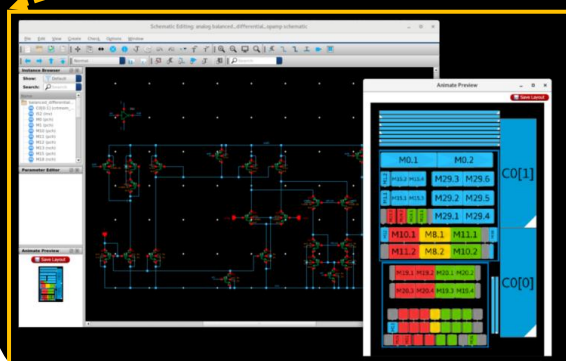
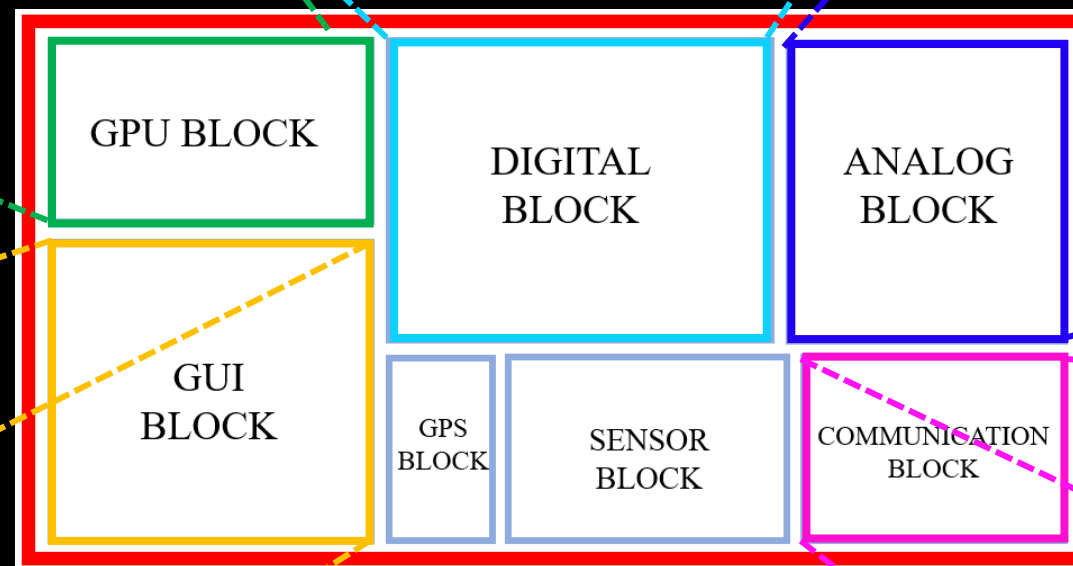
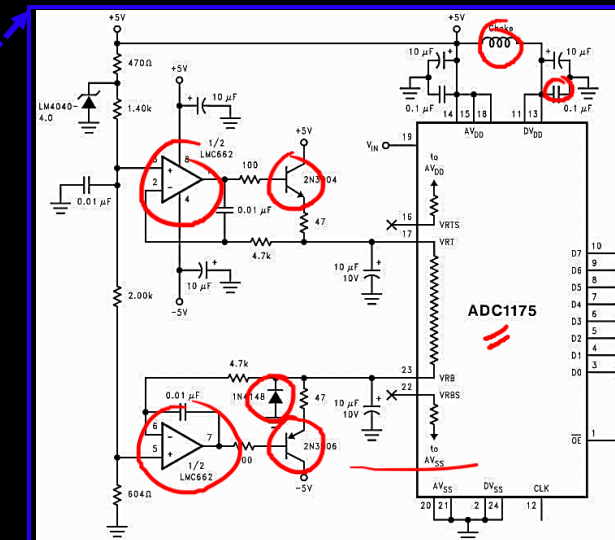
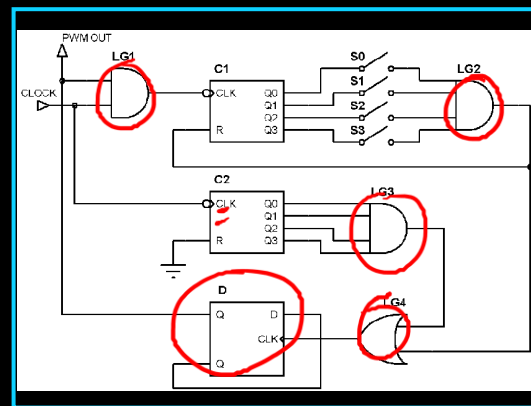
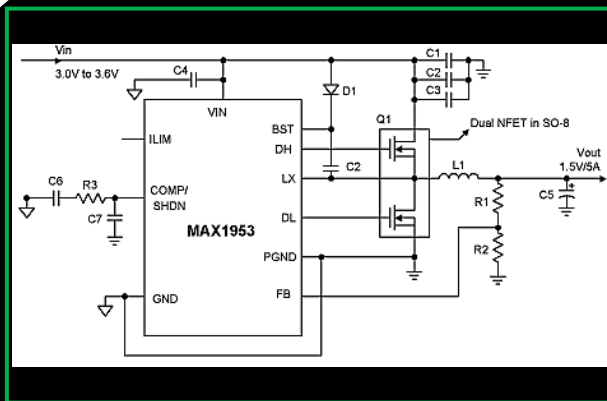
- I. ADC (Analog-to-Digital Converter)
- II. DAC (Digital-to-Analog Converter)
- III. Low-Dropout Regulators (LDOs)
- IV. Oscillator (RC/Crystal)
- V. Bandgap Reference
- VI. Temperature Sensor
- VII. Analog MUX
- VIII. Charge Pump
- IX. Voltage Comparator
- X. Audio Codec (ADC/DAC Pair)



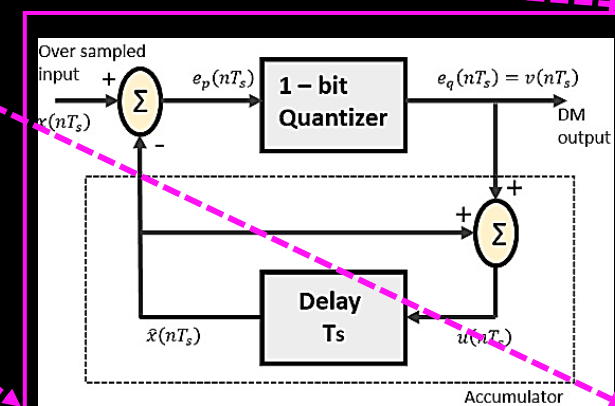
- I. Display Buffer
- II. UI Event Handler
- III. Touch Controller Interface
- IV. Font/Bitmap Engine
- V. Animation Engine
- VI. Frame Scheduler
- VII. Interrupt Controller
- VIII. DMA Engine
- IX. Color Converter
- X. Display Refresh Controller

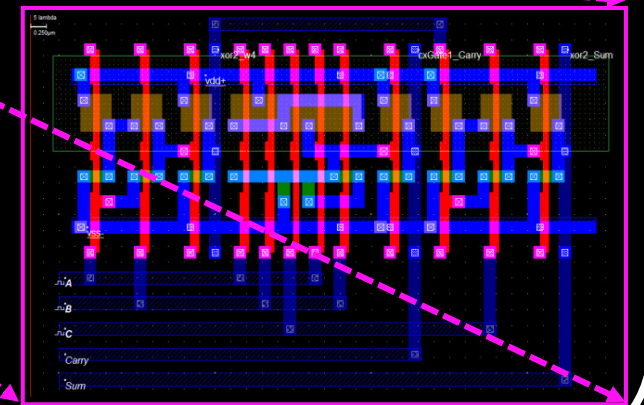
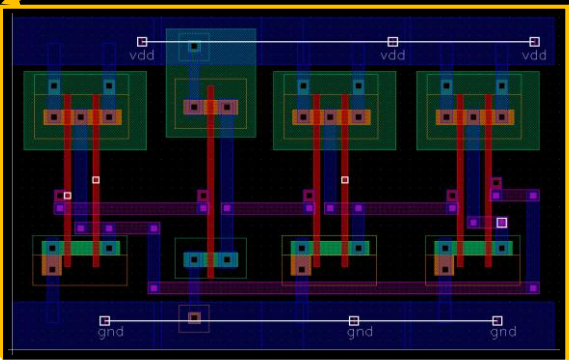
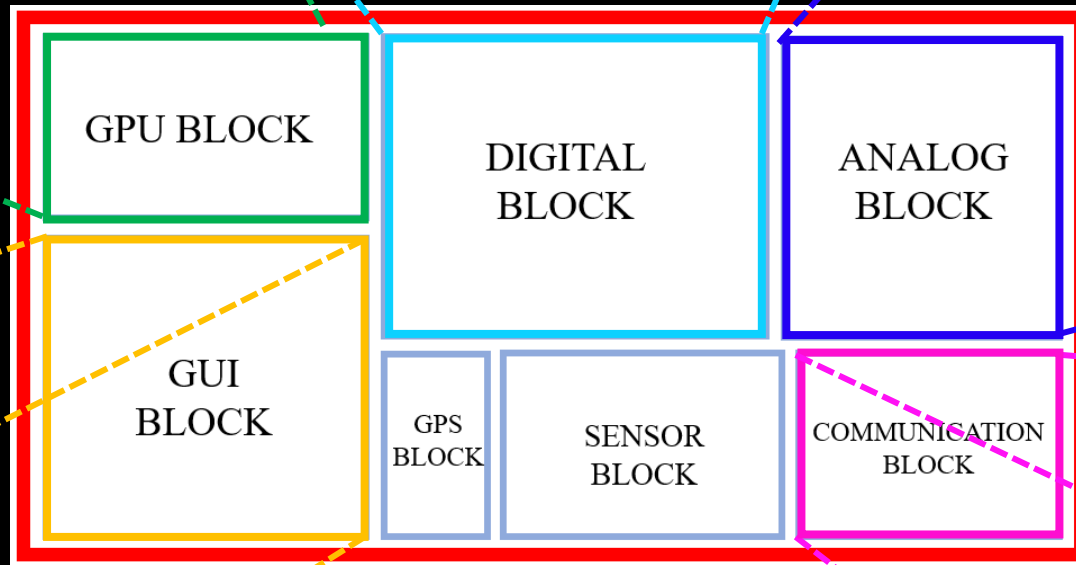
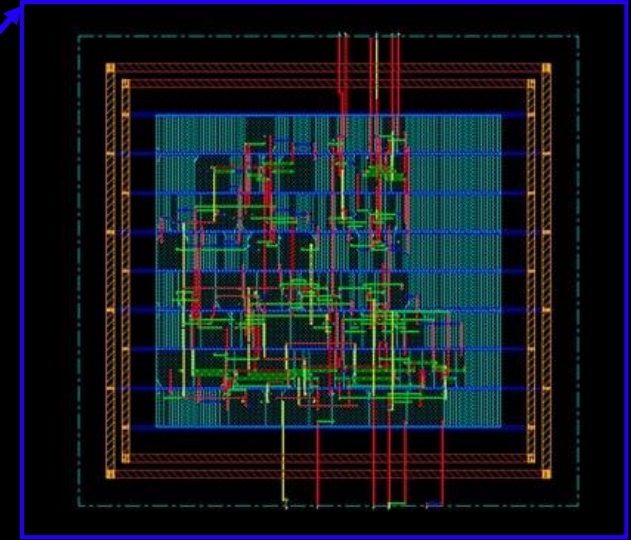
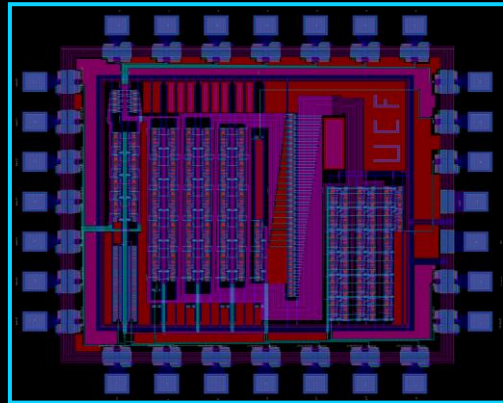
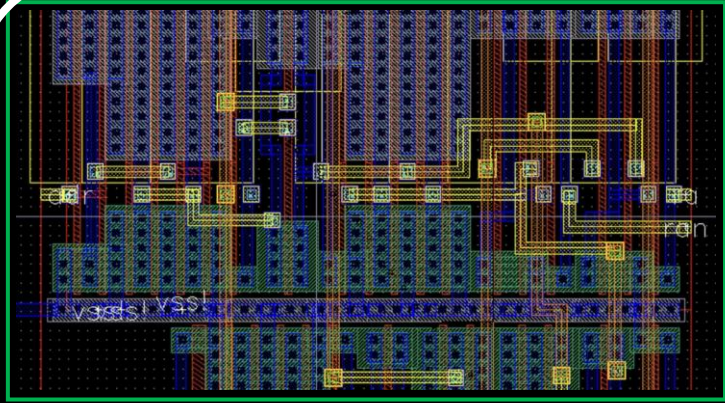
III. FUNCTIONAL DESIGN

- I. Bluetooth Transceiver
- II. Wi-Fi Transceiver
- III. LTE Modem
- IV. RF Front-End
- V. Baseband Processor
- VI. MAC Layer Controller
- VII. Antenna Switch Module
- VIII. Clock Synthesizer
- IX. Power Amplifier
- X. Low Noise Amplifier (LNA)



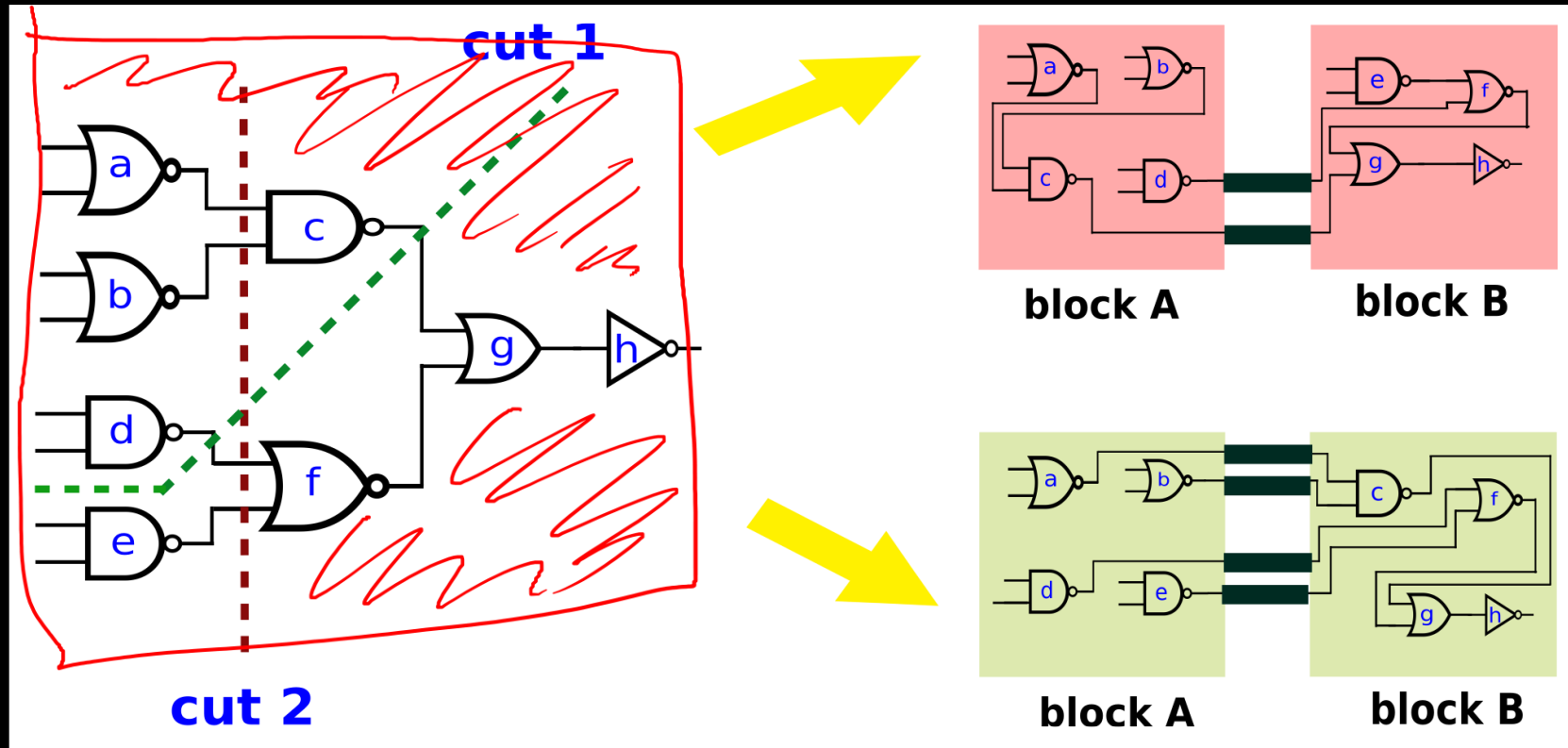
IV. LOGIC/CIRCUIT DESIGN





V. PHYSICAL DESIGN (Layout)

V.a. Partitioning



V.b. Floorplanning

